

Creditor Rights, Technology Adoption, and Productivity: Plant-Level Evidence

Nuri Ersahin

Michigan State University

I use U.S. Census microdata to analyze the effect of stronger creditor rights on productivity. Following the adoption of antirecharacterization laws that give lenders greater access to the collateral of firms in financial distress, treated plants' total factor productivity increases by 2.6%. This effect is concentrated among plants belonging to financially constrained firms. I explore the underlying mechanism and find that treated plants change the composition of their investments and their workforce toward newer capital and skilled labor. My results suggest that stronger creditor rights relax borrowing constraints and help firms adopt more efficient production technologies. (JEL D24, G32, G33, K22)

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An influential body of research argues that legal institutions affect the economic organization of firms and the growth of an economy (King and Levine 1993; La Porta et al. 1997; Rajan and Zingales 1998). Although much has been learned, we still know little about the mechanisms by which specific legal rules and features govern firm financial and real outcomes.

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For example, the extent to which certain legal institutions lead to a relaxing of financial constraints and a consequent improvement in productivity via technological adoption and/or higher levels of investment is unclear. In this paper, I attempt to establish evidence along these lines by considering the role of secured creditor rights in bankruptcy on firm productivity and production processes, including the composition of both investment and workforce.

Stronger creditor rights, such as better legal enforcement, is a critical determinant of firm financing (Hart and Moore 1994; Rampini and Viswanathan 2013). More specifically, expansion of the menu of collateralizable assets or increased ability to recover collateral in bankruptcy may reduce the costs of lending for creditors, thereby leading to a lower cost of external financing for firms (Haselmann et al. 2010; Cerqueiro et al. 2016; Gopalan et al. 2016; Calomiris et al. 2017). With broader access to credit, firms can increase both their investment and employment (Campello and Larrain 2016; Gopalan et al. 2016; Aretz et al. forthcoming).

However, creditor rights can generate differences in not only the *quantity* but also the *quality* of investments and workforce and, consequently, productivity. Regarding investment, Eisfeldt and Rampini (2007) and Rampini (2019) argue that financial constraints constitute the main reason firms invest in used rather than new capital. Therefore, assuming that newer goods are more productive because of technological progress, relaxation in borrowing constraints might lead to higher technology adoption by firms, which could help them operate more efficiently (Midrigan and Xu 2014). Regarding workforce, Griliches (1969) and Krusell et al. (2000) argue that equipment and information technology (IT) capital are more complementary to skilled labor than to unskilled labor. According to this capital-skill complementarity hypothesis, change in the composition of investment should lead to a change in the composition of workforce by increasing the demand for skilled labor.

Two obstacles hinder any attempt to study the effect of creditor rights on both firms' productivity and composition of investment and workforce. First, standard corporate data sources generally report only aggregate items, such as capital expenditures or the number of employees, which does not allow a deeper look into the constituents. The disaggregated nature of the Census data reveals the type of capital and labor a firm invests in. The second obstacle relates to endogeneity: a variety of unobservable factors may be affecting both creditor rights in a country or state and firm behavior. To address this concern, I use enactment of antirecharacterization statutes as a source of exogenous variation in creditor rights (e.g., Li et al. 2016). These laws mainly affect the securitization industry and firms using a special purpose vehicle (SPV) to conduct secured borrowing. Collateral is transferred to a SPV for the purpose of protecting it from automatic stay in case of the bankruptcy of the debtor. However, before these laws, bankruptcy judges had the discretion to make the collateral in the SPV subject to automatic stay. These laws significantly increased the rights of

secured creditors in bankruptcy by denying judges this discretion and allowing secured creditors to seize the collateral in the SPV.¹

I adapt the empirical methodology of Bertrand and Mullainathan (2003) to study the effect of antirecharacterization laws based on plant-level data from the U.S. Census Bureau. The granularity of the Census data coupled with the fact that antirecharacterization laws were adapted at the state of incorporation level helps me conduct two types of analysis. First, the Census data provide the exact geographic location and industry of each plant, which allows me to compare two plants in the same year, industry, and location. To illustrate, I compare the productivity change of two plants in Iowa, one of which belongs to a firm incorporated in Texas (a treated state) and the other that belongs to a firm incorporated in California (a control state). This plant-level analysis allows me to observe the productive efficiency of a plant far from the state of headquarters or state of incorporation, and separate out the effect of stronger creditor rights from the effect of local economic shocks contemporaneous with the laws. Second, the longitudinal nature of the plant-level data helps me observe the productivity of a plant for several consecutive years, which is crucial in the context of this study, given that effects on productivity may take time to materialize.

I find that following the adoption of antirecharacterization laws, total factor productivity (TFP) of plants belonging to treated firms increases significantly, by 2.6%. Further, with an increased access to financing, long-term and total debt significantly increase at treated firms.

Next, I explore the underlying mechanism through which strengthening of creditor rights and the resultant increased access to financing leads to greater productivity. To this end, I explore the unique features of the Census data and look at the composition of investments and workforce. First, I show that treated plants adopt more advanced technology by investing in newer capital and IT; new machinery and computer investments significantly increase at treated plants. Consistent with Eisfeldt and Rampini (2007), who show that investing in used capital rather than new capital is more common among credit-constrained firms, I document that the fraction of capital expenditures in new capital, new machinery, and IT significantly increases at treated plants.² Changing composition of investment is related to financial constraints: only plants belonging to financially constrained firms experience an increase in both the fraction of capital expenditures on new capital and productivity. This

¹ A substantial number of firms use SPVs. Of 6,473 public firms between 1997 and 2004, Feng et al. (2009) find that on average, 42% of them use at least one SPV. In 2004, that 59% percent of firms reported at least one SPV also highlights their prevalence (Feng et al. 2009). Further, Korgaonkar and Nini (2010) state that firms in manufacturing industries intensively rely on SPVs.

² Using data from the Annual Capital Expenditure Survey (ACES) conducted by the U.S. Census Bureau, Eisfeldt and Rampini (2007) show that the fraction of capital expenditures on used capital is 27.79% for firms in the lowest size decile, with assets below \$0.10 million, and 10.10% for firms in the highest asset decile, with assets above \$186.55 million. Using other measures of credit constraints, they report that this fraction increases significantly as firms become more constrained.

finding fits well with that of Midrigan and Xu (2014), who argue that financial constraints may reduce the productivity of individual producers by distorting their technology-adoption decisions.

Next, to shed further light on the underlying mechanism, I analyze the importance of asset redeployability. Eisfeldt and Rampini (2009) argue that leasing allows higher debt capacity for financially constrained firms because U.S. bankruptcy laws make repossession of assets easier for lessors than for secured lenders. In line with this reasoning, antirecharacterization laws that make it easier for secured creditors to repossess assets in bankruptcy can lead firms to switch from leased (Chu forthcoming) or used assets to new assets. However, not all assets have an active leasing or secondary market. Gavazza (2010, 2011) argues that asset redeployability is a crucial determinant of the thickness of the secondary markets. Therefore, changing composition of investments toward newer assets should be more pronounced for more redeployable assets for which there is an active secondary market. The findings support this hypothesis: using various redeployability measures, I find that plants operating in industries characterized by assets that are more redeployable experience a significant increase in the fraction invested in new capital and new machinery.

Finally, I look at the composition of workforce. The Census provides information about different types of employees, including production workers and nonproduction personnel consisting mainly of supervisors and white-collar employees in sales and marketing, financing, purchasing, and professional and technical roles. I show that total employment increases at treated plants. However, a closer look at different types of employees reveals that only the number of nonproduction workers significantly increases at treated plants. I see no significant change in the number of production workers. This shows that consistent with Griliches (1969) and Krusell et al. (2000), treated plants that change the composition of their investments toward newer and IT capital also change the composition of their workforce by hiring employees that are more skilled.

My findings contribute to the literature on creditor rights and optimal bankruptcy design. Strengthening of creditor rights can result in inefficient liquidation of firms by increasing the cost of continuing during financial distress (Aghion and Bolton 1992; Hart et al. 1999). This liquidation bias can affect ex ante financing and investment policies. Acharya and Subramanian (2009); Acharya et al. (2011); Berger et al. (2011); Seifert and Gonenc (2012); and Vig (2013) present evidence consistent with this view, showing that stronger creditor rights are associated with less innovation, as well as financing policies that are more conservative. A more recent set of studies shows effects contrary to the liquidation bias. Li et al. (2016) show that firms increase leverage significantly after the adoption of antirecharacterization laws. Favara et al. (2019) and Ponticelli and Alencar (2016) document increases in borrowing and investment when creditor rights are stronger. Finally, Mann (2018) and

Cerqueiro et al. (2017) report a positive link between creditor rights and innovation output. I extend this prior work in two ways: First, there is little evidence on the productivity consequences of these creditor rights related financing and investment decisions.^{3,4} Second, I provide new evidence that creditor rights including legal enforcement of financing affect not only the *level* or *quantity* but also the *composition* or *quality* of investments and workforce.⁵

In a closely related paper, Benmelech and Bergman (2011) show that airlines operating in countries with higher creditor protection use aircraft of a younger vintage and newer technology. I extend their work in three different ways. First, I identify a specific mechanism, rights of secured creditors in bankruptcy, and add to our understanding of which aspects of creditor rights matter more for financial and real outcomes (Calomiris et al. 2016). Second, exploiting the industry-level variation of Census data, I uncover the importance of asset redeployability in the relationship between creditor rights and investment. The fact that U.S. bankruptcy laws make repossession of assets easier for lessors than for secured lenders makes secondary markets, as well asset liquidity/redeployability, an important channel through which stronger creditor rights translate into changing composition of investments.⁶ Finally, I complement their work by showing that creditor rights affect both the quality of a firm's capital and the quality of its workforce.

Finally, my study extends a set of studies looking at the effect of finance on different types of labor. Brown and Matsa (2016) and Ramin et al. (2018) show that financial distress prevents firms from hiring and retaining skilled labor. Caggese et al. (forthcoming) argue that financing constraints lead firms to fire short-tenured workers with high future expected productivity. Finally, using industry-level data, Larrain (2014) documents that capital-account opening and the resultant increased access to capital increase sectoral wage inequality by increasing the demand for skilled labor, rather than unskilled labor. I complement these studies by showing that stronger creditor rights can have

³ Productivity shocks have been modeled as an important driver of economic fluctuations in a variety of macroeconomic models, beginning with Kydland and Prescott (1982). Moreover, Hall and Jones (1999) argue that differences in productivity are critical to understanding output differences between countries.

⁴ Some studies consider the effects of finance on productivity. Butler and Cornaggia (2011) study the productivity of farmers following an exogenous increase in demand for U.S. corn. They find that farmer productivity increases significantly in areas with high levels of bank deposits. Krishnan et al. (2014) show that TFP increases after interstate banking deregulations among the smallest firms in the economy. Finally, Banerjee et al. (2019), using the Million Baht Program, which is a large credit expansion program in Thailand, show that preprogram high-TFP households experience a dramatic increase in profits following the program.

⁵ Rampini (2019, p. 668) states the following: "to the best of our knowledge, this basic prediction about the relation between durability and financing has not been directly tested to date. Nor have the predictions regarding the composition of investment in terms of durability across economies with different legal enforcement been investigated empirically."

⁶ Ease of collateral repossession is crucial not only for firms but also for consumers. Using auto loan data from Brazil, Assunção et al. (2013) show that after a credit reform that made it easier for creditors to sell repossessed cars, financially constrained consumers could borrow more and buy newer and more expensive cars. I complement their work by documenting similar evidence but on the corporate side.

differential ex ante effects on different types of labor through a relaxation of borrowing constraints and change in the composition of investments.

1. Antirecharacterization Laws

Antirecharacterization laws affect the securitization industry and firms using a SPV to conduct secured borrowing. The originating firm transfers the collateral to a SPV, which is a financial intermediary designed to be bankruptcy remote, and then sells it to investors as a security. The main reason for transferring assets to a different entity is to characterize them as sales, thereby protecting them from automatic stay in case of bankruptcy of the issuing firm. The bankruptcy remoteness of a SPV assures investors that obligations will be fulfilled even if the SPV-originating firm goes bankrupt. However, the bankruptcy remoteness of a SPV should not be taken for granted. Sometimes bankruptcy courts can recharacterize the asset transfer as a secured loan rather than as a true sale, making the SPV assets subject to automatic stay. Bankruptcy courts generally justify this by stating that the SPV will play an important role in the reorganization of the bankrupt firm. The enactment of antirecharacterization laws discards the possibility of recharacterization by bankruptcy courts.

Seven states enacted these laws: Texas and Louisiana in 1997, Alabama in 2001, Delaware in 2002, South Dakota in 2003, Virginia in 2004, and Nevada in 2005. Kettering (2010) states that these seven states fall into two groups in terms of the coverage of their antirecharacterization laws. Texas and Louisiana use Section 9.109 of the Uniform Commercial Code (UCC) to discard the possibility of recharacterization for all sales of receivables, whereas the remaining states use stand-alone statutes to discard this possibility only for sales under a securitization transaction.⁷

Although seven states enacted these laws, I consider only Texas, Louisiana, and Alabama as the treatment states because of the 2003 federal court ruling on *Reaves Brokerage Company, Inc., v. Sunbelt Fruit & Vegetable Company, Inc.* Reaves, a fresh food and vegetable seller, was making frequent sales to Sunbelt, a wholesaler. Sunbelt ceased operations in March 2000, owing Reaves \$195,060.55. Reaves sought recovery from Sunbelt under the Perishable Agricultural Commodities Act (PACA). A couple of months later, Reaves sought recovery also from Fidelity Factors, LLC, a factoring company that had purchased several accounts receivable from Sunbelt. PACA was enacted in 1930 and later amended in 1984 to protect sellers of perishable agricultural products. PACA requires that perishable product sellers be paid promptly and their claims prime those of other secured and unsecured creditors in case the buyer gets out of business. Based on this, Reaves argued that the sale of accounts is not a true sale but just a disguised way of financing, which makes Fidelity a

⁷ Appendix A gives the section of the state statutes that guarantees antirecharacterization for Texas, Louisiana, and Alabama, which constitute the treatment states.

secured creditor whose claims must be subordinate to theirs. Fidelity defended itself based on the Texas antirecharacterization statute, arguing that they are buyers of accounts receivable and not lenders. The court ruled in favor of Reaves, stating that the relationship between Fidelity and Sunbelt was that of a secured lender and debtor, not that of a seller and buyer. This ruling created uncertainty around antirecharacterization laws because it created a precedent where federal courts could overrule state antirecharacterization laws.⁸

A deeper look at antirecharacterization laws shows that these laws affect a substantial number of transactions. Section 9.109 of the Texas and Louisiana UCC explicitly discards the possibility of recharacterization of the sale of the following four items: accounts, chattel paper, payment intangibles, and promissory notes.⁹ The most important item for the purposes of the study is accounts, defined in Section 9.102 of the UCC. *Accounts* means a right to payment of a monetary obligation, whether or not earned by performance, for a variety of items, including (a) property that has been or is to be sold, leased, licensed, assigned, or otherwise disposed of; (b) services rendered or to be rendered; (c) a policy of insurance; (d) licensing of intellectual property; and (e) credit cards. This definition extends beyond the traditional definition of *accounts* consisting mainly of trade receivables.

In Alabama, the third treatment state, and in the remaining four states, an act that precludes recharacterization of a sale of property of any kind included in a securitization transaction guarantees antirecharacterization. In Appendix A, Section 35-10A-2 of the Code of Alabama states that no property, assets, or rights transferred to a SPV can be subject to recharacterization. Assets typically transferred to a SPV include trade receivables, residential and commercial loans, equipment leases, licenses, and management contracts.

Taken together, these laws increase the pledgeability of any rights to future payment (i.e., receivables) by making them more accessible to creditors. Manufacturing firms, which constitute the sample, in particular generate a large amount of receivables. Korgaonkar and Nini (2010) state that firms involved in the manufacturing and production of consumer durables are heavy users of SPVs. For this reason, I expect these antirecharacterization laws to have a significant effect on manufacturing firms.

⁸ Li et al. (2016) document that sixty-two other bankruptcy cases in the following 7 years cited *Reaves Brokerage Company, Inc., v. Sunbelt Fruit & Vegetable Company, Inc.* as a precedent. Karpoff and Wittry (2018) validate the importance of this federal court ruling. They show that important court rulings must be taken into account when analyzing the incremental impact of antitakeover laws.

⁹ Part of Section 9.109 of the Texas and Louisiana UCC that explicitly discards the recharacterization: "The application of this chapter to the sale of accounts, chattel paper, payment intangibles, or promissory notes is not to recharacterize that sale as a transaction to secure indebtedness but to protect purchasers of those assets by providing a notice filing system. For all purposes, in the absence of fraud or intentional misrepresentation, the parties' characterization of a transaction as a sale of such assets shall be conclusive that the transaction is a sale and is not a secured transaction and that title, legal and equitable, has passed to the party characterized as the purchaser of those assets regardless of whether the secured party has any recourse against the debtor, whether the debtor is entitled to any surplus, or any other term of the parties' agreement."

Manufacturing firms are not the only users of securitization. Feng et al. (2009) document that the percentage of firms using at least one SPV increased from 23% in 1997 to 59% in 2004, which confirms how common they became in the period the sample covers. Finally, these laws are fairly exogenous to the firms included in the sample. Kettering (2008) describes the enactment of antirecharacterization laws as a great success of the securitization industry. Janger (2003) constitutes another study regarding the big role that the financial industry played in the enactment of antirecharacterization laws. I will return to the endogeneity concern that is attributable to lobbying by industrial firms in Section 3.1.1.

2. Data and Empirical Methodology

2.1 Data sources

The main data I use in this study are a combination of two data sets from the Census. Both data sets cover only U.S. manufacturing plants. The first manufacturing plant data set is the Census of Manufacturers (CMF). The CMF is a survey conducted every 5 years, which consists of all manufacturing plants in the United States with at least one paid employee. The second manufacturing data set is the Annual Survey of Manufacturers (ASM). The ASM is conducted in years when the CMF is not conducted, and it includes a subset of plants surveyed in the CMF. It includes all plants with more than 250 employees and some with fewer employees, included with a probability positively correlated with their number of employees. Both of these data sets include detailed information on industry, corporate affiliation, total shipments, employment, capital expenditures, and material inputs of each plant. Reporting for both of these surveys is mandatory, and misreporting is penalized, which alleviates misreporting and response rate concerns. The level of granularity of these manufacturing data sets plays a critical role in the analysis. First, it allows for the construction of various measures of productivity for each manufacturing plant, and thereby, for analysis of how the productivity of a plant and the creditor rights are related. Second, it makes it possible to study different types of capital and labor by providing granular and reliable machinery, computer, and employment numbers.

In addition to these two manufacturing data sets, I use another data set from the Census, the Longitudinal Business Database (LBD). The LBD keeps yearly track of all business establishments in the United States with at least one paid employee. It provides information on the number of employees, payroll, geographical location, industry, and corporate affiliation for each establishment. For the purpose of this study, I use the LBD to get information about the age of plants and the number of plants firms own.

Finally, I use firm-level data from Compustat. I confine the analysis to public firms because I can observe the state of incorporation only for public firms. To match firms in Compustat to plants, I use the Compustat bridge of the Census

Bureau that ends in 2005. I extend the bridge to 2009 by making use of various employer characteristics, including name, address, and identification number (EIN). The sample covers the period from 1992 to 2009.

2.2 Variable construction and summary statistics

I consider firms headquartered or incorporated in Texas or Louisiana between 1997 and 2003, or Alabama between 2001 and 2003 as treated firms. The question of which state law will govern recharacterization is complicated.¹⁰ Article 9 of the UCC specifies the rules governing secured transactions, including securitization. A revised version of Article 9, effective as of July 2001, states that the law governing a secured party's interest in receivables is the law of the location of the debtor, which is defined to be the state in which the entity is registered. For corporations, this definition of location of debtor means the state of incorporation.¹¹ However, the old version of Article 9, effective before June 2001, deemed the debtor's location to be the location of its chief executive office, that is, the headquarters. The official comment to this older version of Article 9 explicitly states that the chief executive office is not necessarily the place of incorporation but is the place from which the debtor manages the main part of his business operations. Therefore, both the state in which the headquarters is located and the state of incorporation must be taken into account.¹²

The main dependent variable is the natural logarithm of total factor productivity, which I construct at the plant level following the methodology in Foster et al. (2016).

$$TFP_{it} = \ln Q_{it} - \alpha_{kt} \ln K_{it} - \alpha_{lt} \ln L_{it} - \alpha_{mt} \ln M_{it}, \quad (1)$$

where i and t index for plant and year, respectively. The variables TFP , Q , K , L , M , and α represent total factor productivity, real output, capital stock, labor input, cost of materials and parts, and factor elasticities, respectively.¹³ I construct output by summing the total value of shipments and change in inventories for finished goods and work-in-process. Because the Census does not collect establishment-level prices, I use an industry-level measure from the NBER-CES Manufacturing Industry Database to deflate output.¹⁴

¹⁰ See Kettering (2010) for an extensive analysis of the choice of law governing recharacterization.

¹¹ Parties to secured loans cannot choose which state law will govern the ownership of collateral in bankruptcy. *In re Eagle Enterprises* constitutes the benchmark case for this. Please see Mann (2015) for more details.

¹² See Appendix B for the choice of law rules specified in both old and new versions of Article 9.

¹³ See Foster et al. (2016) for a more detailed explanation of how I construct each variable.

¹⁴ One concern with using a common industry-level deflator is that TFP differences across producers may not represent their true technological efficiency differences in the case of idiosyncratic demand shocks or market power variation within an industry. Foster et al. (2008) partially alleviate this concern by finding a high correlation (0.75) between revenue- and quantity-based productivity measures for the set of industries for which they had both plant-level price and quantity data.

Using a perpetual inventory method, I construct capital stock that consists of structure and equipment. I take a plant's first year in the CMF/ASM records as a starting point to construct the capital stock series. I then add capital expenditures, using investment price deflators from the Bureau of Labor Statistics (BLS) at the 2-digit SIC or 3-digit NAICS level, to the capital stock each year. I use an industry-level depreciation rate from the Bureau of Economic Analysis. I construct the capital stock series for structure and equipment separately. They sum to the total capital stock owned by the plant. I use "production worker equivalent hours" as the measure of labor input, which I construct by multiplying the number of hours worked by production workers with total wages divided by wages paid to production workers. Materials costs include both non energy and energy-related costs. Nonenergy materials costs include costs of materials and parts, as well as cost of resales and contract work. Energy costs include electricity and fuel costs. I deflate both types of costs by their corresponding industry deflators. Finally, I use industry-level factor cost shares for factor elasticities from the NBER Productivity Database.

In addition to TFP, I analyze the effects of stronger creditor rights on composition of investment and workforce. I use two new variables for investment: *machinery investment* and *computer investment*. I construct *machinery investment* by scaling machinery expenditures by lagged machinery and equipment stock. To represent investment in IT, I use the *capital expenditures for computers* variable from the CMF and ASM databases and scale it by lagged machinery and equipment stock. Specifically, *capital expenditures for computers* includes computer hardware, software, and network expenditures. I have this variable starting with 2000. As to composition of investment, the Census data allow me to see capital expenditures in new and used goods, separately. I define four new variables available until 2001: *Fraction New Capital*, new capital expenditures over total capital expenditures; *Fraction New Machinery-1*, new machinery expenditures over total machinery expenditures; *Fraction New Machinery-2*, new machinery expenditures over total capital expenditures; and *Fraction computer*, capital expenditures for computers expenditures over total capital expenditures.

Alongside composition of investment, I also analyze composition of workforce. Census microdata provide information about two types of employees: production workers and nonproduction personnel. Production workers consist of employees engaged in production operations at the plant up through the working foreman level. Nonproduction personnel consist of supervisors that are above the foreman level and office employees. To study the growth rate of these two types of labor, I calculate the symmetric growth rate of employment as two times the change in the number of employees scaled by the sum of the current and lagged number of employees. A big advantage of this measure is that it limits the effects of extreme values (Davis et al. 1998). To study the composition of workforce, I define *Fraction nonproduction emp.* as the natural logarithm of the number of nonproduction employees over the

total number of employees. I winsorize all variables at 1% to ensure that the results are robust to outliers.

Table 1 presents summary statistics for the aggregate sample and the subsample of plant-year observations treated by antirecharacterization laws.¹⁵ The final sample contains 220,000 plant-year observations covering the period 1992–2009.¹⁶ The subsample of treatment plants accounts for a relatively small fraction of the total plant-year observations. However, the treatment subsample nearing 10,000 plant-year observations is sufficient, given that the treatment effect requires only that the treatment subsample be sufficiently large in absolute terms, as argued in Giroud (2013). Control plants are older and bigger, which constitutes one of the main reasons I include the age of plants and their total value of shipments to control for differences between treatment and control plants.

2.3 Identification and empirical model

I follow an “on-off” approach in the estimation strategy because of the 2003 federal court ruling. According to this approach, the “on” period includes years after a state passes antirecharacterization laws, but before 2004. I do not include 2004 and the following years because of the 2003 federal court ruling that created substantial uncertainty around these laws. To be more specific about the treatment variable, for a firm incorporated or headquartered in Texas or Louisiana, the “on” period consists of 1997 to 2003, whereas the “off” period consists of 1992 to 1996 and 2004 to 2009. For a firm incorporated or headquartered in Alabama, the “on” period consists of 2001 to 2003, whereas the “off” period consists of 1992 to 2000 and 2004 to 2009.¹⁷ I estimate the following equation:

$$y_{ifhklst} = \alpha_{lst} + \alpha_i + \alpha_h + \alpha_k + \gamma X_{ifhklst} + \delta ARL_{hkt} + \epsilon_{ifhklst}, \quad (2)$$

where i , f , h , k , l , s , and t index for plant, firm, state of headquarters, state of incorporation, state of location, industry, and year, respectively. The unit of observation is a plant-year pair. The variable y stands for variables related to productivity, composition of investment, and workforce. ARL_{hkt} is an indicator variable that equals one if the plant belongs to a firm whose state of

¹⁵ Following the disclosure requirements of the Census, I do not report quantile values; I round off the number of observations in each table.

¹⁶ In Appendix IA.I, I reestimate the baseline model, considering the end year to be 2003. Results are similar to the baseline.

¹⁷ I conduct several tests to ensure that the results are not an artifact of this estimation approach. First, in Appendix IA.I, I adopt a “before-after” approach by restricting the time period to 1992–2003. Second, I conduct two more tests in Appendix IA.II. In Column 1, I exclude all firms incorporated in states that enacted antirecharacterization laws after the 2003 federal court ruling, that is, South Dakota, Virginia, and Nevada. In Column 2, I adopt another “before-after” approach by ignoring the 2003 federal court ruling and considering all seven states enacting antirecharacterization laws, Texas, Louisiana, Alabama, Delaware, South Dakota, Virginia, and Nevada, to be treated states. The baseline results are robust to all of these tests.

Table 1
Creditor rights and productivity: Summary statistics

	Full sample			Recharacterization			Antirecharacterization		
	Rounded <i>N</i>	Mean	SD	Rounded <i>N</i>	Mean	SD	Rounded <i>N</i>	Mean	SD
Total payroll	220,000	15,170	46,812	210,000	15,284	47,254	10,000	11,885	31,233
Production payroll	220,000	8,457	22,707	210,000	8,484	22,748	10,000	7,662	21,462
Employees	220,000	339	787	210,000	342	795,609	10,000	250	482
Total hours	220,000	838	2,177	210,000	846	2196	10,000	617	1499
Average wage	220,000	39,328	17,631	210,000	39,337	17,689	10,000	39,079	15,860
Labor productivity	220,000	3,955	0,893	210,000	3,952	0,891	10,000	4,051	0,931
Total factor productivity	220,000	1,802	0,593	210,000	1,802	0,594	10,000	1,822	0,577
Output	220,000	150,495	1,633,717	210,000	148,267	1,629,079	10,000	215,044	1,761,671
Plant size	220,000	10,531	1,472	210,000	10,542	1,463	10,000	10,220	1,687
Plant age	220,000	19,083	8,825	210,000	19,130	8,821	10,000	17,709	8,821
Return on capital	220,000	1,481	2,083	210,000	1,482	2,081	10,000	1,434	2,136
Capital stock	220,000	55,711	255,295	210,000	55,265	256,634	10,000	68,622	212,485
Employment growth	170,000	-0,026	0,224	165,000	-0,026	0,090	5,000	-0,027	0,254
Total investment	118,000	0,096	0,129	113,000	0,096	0,128	5,000	0,103	0,142
Machinery investment	118,000	0,137	0,174	113,000	0,137	0,174	5,000	0,138	0,189
Computer investment	90,000	0,005	0,012	87,000	0,005	0,012	3,000	0,004	0,011
Total employment	108,000	407	862	103,000	410	871	5,000	328	540
Production employment	108,000	280	512	103,000	281	514	5,000	239	431
Nonproduction employment	108,000	127	479	103,000	129	482	5,000	89	192

Plant-level data are taken from the CMF/ASM-Compustat match for the years 1992–2009. The unit of observation is a plant-year pair. Table A1 in the appendix defines all variables.

headquarters (h) or state of incorporation (k) passed an antirecharacterization law by year (t) and by year 2003. To illustrate, ARL will equal zero in year 2007 for a plant belonging to a Nevada-incorporated firm, because Nevada passed an antirecharacterization law in 2005, after the 2003 federal court ruling. α_{lst} , α_i , α_h , and α_k are state of location $\times$ industry $\times$ year, plant, state of headquarters, and state of incorporation fixed effects, respectively. $X_{ifhklst}$ stands for plant and firm controls: total value of shipments, age of the plant, and number of plants owned by the firm the plant belongs to. I define *Age* based on a plant's first year in LBD records. I categorize industries at the 2-digit SIC level. I cluster standard errors at the state of location level.¹⁸

I control for time-invariant characteristics at the plant level through plant fixed effects. I use state of incorporation fixed effects to control for time-invariant differences between treated and untreated plants. In addition, I use state of headquarters fixed effects, given that the definition of treatment includes both state of incorporation and state of headquarters. The state of location dummies interacted with industry and year dummies, α_{lst} , allow me to control for contemporaneous shocks at the state of location and industry levels. I use state and industry dummies for two main reasons. First, TFP shows considerable variation across industries (Foster et al. 2016). Second, the TFP measure is essentially a revenue measure. Demand shocks at different geographies may inflate or deflate the TFP measure in a way unrelated to the productive efficiency, which constitutes the main reason to control for state of location. For these two main reasons, I use state $\times$ industry $\times$ year fixed effects to control for contemporaneous shocks at the state of location level. I use fixed effects instead of demeaning the dependent variable because demeaning can lead to inconsistent estimates (Gormley and Matsa 2013).

The coefficient of interest is δ , which measures the effect of antirecharacterization laws on TFP, composition of investment, and composition of workforce. δ measures the difference between the productivity change of two plants located in the same state (e.g., Iowa) and operating in the same industry (e.g., food manufacturing), one of which belongs to a firm incorporated in Texas (a treated state), and the other that belongs to a firm incorporated in California (a control state).

3. Empirical Results

3.1 Antirecharacterization laws and productivity

I first estimate Equation (2) with the natural logarithm of TFP as the dependent variable.¹⁹ Table 2 presents the results.

¹⁸ In regressions not reported here, I cluster standard errors at the state of headquarters, state of incorporation, and firm level and find similar results.

¹⁹ In Appendix IA.III, I estimate Equation (2) with standardized TFP as the dependent variable. I scale TFP by its standard deviation to standardize. Estimated coefficients indicate an increase in TFP ranging between 4.4% and 5.3% of its standard deviation among the full sample of plants.

Table 2
Creditor rights and productivity: Baseline estimates

Dependent variable: *TFP*

	[1]	[2]	[3]
<i>ARL</i>	0.032*** (0.008)	0.027*** (0.010)	0.026*** (0.010)
Control variables	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
State of incorporation fixed effects	No	Yes	Yes
State of headquarters fixed effects	No	Yes	Yes
State of location $\times$ Industry $\times$ Year fixed effects	No	No	Yes
Rounded <i>N</i>	220,000	220,000	220,000
<i>R</i> ²	.77	.77	.74

This table presents estimates of the plant-level effect of antirecharacterization laws on total factor productivity (TFP). The main independent variable is *ARL*, which is an indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2003 or Alabama between 2001 and 2003. The unit of observation in each regression is a plant-year pair. Control variables include the size and age of the plant and the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Column 1 presents results from estimation of Equation (2) with only plant and year fixed effects. The coefficient of interest on *ARL* is 0.032, and is significant at the 1% confidence level. This estimate corresponds to a 3.2% increase in TFP, which is in line with the prediction that stronger creditor rights will lead firms to operate more efficiently.

Column 2 adds state of incorporation and state of headquarters fixed effects to control for time-invariant differences between treated and untreated firms. The inclusion of these fixed effects decreases the estimated coefficient to 0.027. However, this coefficient is still significant at the 1% confidence level, and it translates into a 2.7% increase in TFP.

Column 3 adds state of location-industry-year fixed effects to the estimation in Column 2 to tighten the specification. As discussed in Section 2.3, these interacted dummies allow me to control for contemporaneous shocks at the state-of-location level. The inclusion of these controls makes a minor change to the estimated coefficient of interest, which is 0.026 and significant at the 1% confidence level. To put this estimate in economic terms, a 2.6% increase in TFP corresponds to a 2.6% increase in revenues with the same set of inputs (i.e., holding costs constant). Because profits equal revenues minus costs, the percentage increase in profits will depend on the profit margin. Given that the pretreatment average profit margin is 35%, a 2.6% increase in revenues will lead to a $2.6 \times (100/35) = 7.43\%$ increase in profits, which is economically significant.^{20,21}

²⁰ I use an example very similar to the one in Krishnan et al. (2014) to clarify this calculation. Assume that the cost is \$100 and revenue is \$125, which makes the profit margin (profits over revenues) equal 20% ($25/125 = 0.2$). A TFP increase of 2.6% means that revenue increases by 2.6%, whereas cost stays constant. Revenue becomes \$128.25 (125×1.026), and profit becomes \$28.25 ($128.25 - 100$). This shows that given a 20% margin, a 2.6% increase in TFP translates into a 13% ($2.6 \times (100/20) = 13$) increase in profit ($(28.25 - 25)/25 = 0.13$).

²¹ This 7.43% increase in profit is likely an overstatement, because plant-level data do not include overhead costs related to administrative headquarters or off-site marketing and sales departments.

Taken together, these estimates indicate that the passage of antirecharacterization laws has an economically large and statistically robust effect on TFP of treated plants. These estimates suggest that stronger creditor rights have a positive effect, leading firms to operate more efficiently.

3.1.1 Robustness checks. Next, I conduct several tests to ensure the robustness of the results in Table 2. First, in Appendix IA.IV, I estimate two alternative specifications. Specification 1, in Columns 1 to 3, includes a *Post-ARL* dummy equaling one for Texas and Louisiana from 1997 onward, and one for Alabama from 2001 onward, and a separate *Reaves* dummy equaling one for these three states from 2004 onward. Although estimated coefficients on *Reaves* are not statistically significant, estimated coefficients on *Post-ARL* are close to Appendix IA.I in terms of magnitude, and higher than benchmark estimates. This shows that the effect is mainly coming from the period before the federal court ruling. Specification 2, in Columns 4 to 6, further augments Specification 1 with a separate dummy, *DSVN*, which equals one for the states that pass antirecharacterization laws after the federal court ruling (i.e., *DSVN* equals one for Delaware, South Carolina, Virginia, and Nevada from 2002, 2003, 2004, and 2005 onward, respectively). Specification 2 reinforces the conclusion that the effect is stronger before the federal court ruling.

Next, I further account for heterogeneity between treatment and control samples. First, to control for unobserved time-invariant and time-varying factors at the firm level, I augment Equation (2) with firm fixed effects and firm-level control variables, including *sales*, *profitability*, *Tobin's q*, and *tangibility*. Appendix IA.V shows that benchmark results are robust to this new specification. Second, I adopt a nearest-neighbor propensity score matching scheme. The treatment group consists of firms headquartered or incorporated in Texas, Louisiana, or Alabama. One year before the enactment of antirecharacterization laws, I match each treatment plant in Texas, Louisiana, and Alabama to a control plant using a nearest-neighbor propensity score matching with replacement. I estimate propensity scores based on plant size, plant age, and firm size. Plant state of location and plant industry require an exact match. I also require that both sets of plants have at least one observation before and at least one observation after the enactment of antirecharacterization laws.

Appendix IA.VI displays the results. Panel A tabulates the means of the matched variables for the treatment and control groups 1 year before the enactment of antirecharacterization laws. The sample means of control variables for matched treated and control samples are not significantly different. Panel B presents estimates of the plant-level effect of antirecharacterization laws on TFP using these propensity-score-matched samples from the period 1992–2003. The estimated coefficient of interest is 0.030, which is very close to the benchmark estimate of 0.026.

One potential problem is related to the endogeneity of antirecharacterization laws. If firms lobbied for these laws, then their enactment might be correlated with unobserved factors affecting firms' productivity. First, Kettering (2008) describes the enactment of antirecharacterization laws as a great success of the securitization industry, and argues that these statutes are the product of efforts by the financial industry to abolish the possibility of recharacterization.²² Second, I address this concern with a dynamic version of Equation (2) to investigate when the effects of these laws materialize. If these laws are the result of economic factors leading firms to lobby for them, then I should be able to detect their effect before their enactment.

Table 3 investigates the dynamic effects of antirecharacterization laws. First, the estimated coefficients on $Before^{-2}$ and $Before^{-1}$ are economically small and statistically insignificant. The estimated coefficient on $Before^0$, which represents the year antirecharacterization laws passed, is statistically insignificant. This shows that there is no statistically significant difference in the evolution of productivity at treated and control firms before the enactment of antirecharacterization laws, which is consistent with the parallel trends assumption. Second, the coefficients on $After^1$, $After^2$, and $After^{3+}$ are economically and statistically significant, which shows that the effect of these laws on TFP starts materializing 1 year after their passage.

In Table 4, I conduct another test to check the validity of the results in Table 2 and Table 3. One might worry that regional shocks could affect firms headquartered or incorporated in treated and nearby states. Therefore, the estimates might simply be picking up these regional shocks rather than the effect of antirecharacterization laws. I address this issue in the following way: I falsely assume that states bordering Texas, Louisiana, and Alabama passed the antirecharacterization laws. Estimated coefficients in Columns 1 to 3 are statistically indistinguishable from zero, which shows that the results in Tables 2 and 3 are not artifacts of some regional or political shocks affecting states in the near geography of Texas, Louisiana, and Alabama.²³

A concern with the productivity measure is that it relies on structural assumptions, including the Cobb-Douglas production function. To address this, in Appendix IA.VIII, I use operating margin, labor productivity, and return on capital (ROC) as alternative measures of productivity that do not involve any structural assumptions. I calculate operating margin by scaling total value of shipments minus labor and material costs by total value of shipments. I use the measure of labor productivity used in Brav et al. (2015)

²² Kettering (2008) also reports that the drafters of the Texas antirecharacterization law acknowledged that the law aimed at defending securitization transactions in bankruptcy.

²³ Deregulation of interstate bank branching in the 1990s constitutes an important development in the banking industry that might be driving the results. To address this concern, I augment Equation (2) with *Dereg.*, an indicator variable equaling one in the years after the incorporation state of a firm deregulates interstate bank branching. Appendix IA.VII shows that the estimated coefficient on *Dereg.* is statistically indistinguishable from zero for TFP, composition of investment, and composition of workforce.

Table 3
Creditor rights and productivity: Dynamic estimates

Dependent variable: *TFP*

	[1]	[2]
<i>Before</i> ⁻²	-0.000 (0.008)	-0.000 (0.008)
<i>Before</i> ⁻¹	-0.010 (0.011)	-0.009 (0.011)
<i>Before</i> ⁰	-0.006 (0.012)	-0.011 (0.016)
<i>After</i> ¹	0.041** (0.020)	0.034* (0.020)
<i>After</i> ²	0.037** (0.016)	0.034** (0.017)
<i>After</i> ³⁺	0.033*** (0.009)	0.027*** (0.010)
Control variables	Yes	Yes
Plant fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
State of incorporation fixed effects	No	Yes
State of headquarters fixed effects	No	Yes
State of location × Industry × Year fixed effects	No	Yes
Rounded <i>N</i>	220,000	220,000
<i>R</i> ²	.77	.77

This table presents estimates of the plant-level effect of antirecharacterization laws on total factor productivity (TFP). The unit of observation in each regression is a plant-year pair. *Before*⁻² is an indicator variable that equals one if the plant belongs to a firm headquartered or incorporated in a state that will pass antirecharacterization laws in 2 years. *Before*⁻¹ is an indicator variable that equals one if the plant belongs to a firm headquartered or incorporated in a state that will pass antirecharacterization laws in 1 year. *Before*⁰ is an indicator variable that equals one if the plant belongs to a firm headquartered or incorporated in a state that passes antirecharacterization laws that year. *After*¹ is an indicator variable that equals one if the plant belongs to a firm headquartered or incorporated in a state that passed antirecharacterization laws 1 year ago. *After*² is an indicator variable that equals one if the plant belongs to a firm headquartered or incorporated in a state that passed antirecharacterization laws 2 years ago. *After*³⁺ is an indicator variable that equals one if the plant belongs to a firm headquartered or incorporated in a state that passed antirecharacterization laws 3 years ago or more. Control variables include the size and age of the plant and the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. **p* <.1; ***p* <.05; ****p* <.01.

and Silva (2013): natural logarithm of value added per labor hour, which is the total value of shipments minus material and energy costs divided by the total labor hours. Finally, I define *ROC* as total value of shipments minus labor, material, and energy costs scaled by capital stock. Estimated coefficients of interest are statistically significant at the 1% confidence level in all three columns. Estimated coefficients for labor and capital productivity in Columns 2 and 3 are 0.068 and 0.095, and are economically significant given that they constitute about 7.3% (0.068/0.931) and 4.4% (0.095/2.136) of their standard deviation, respectively. The remarkable increase in labor productivity implies the importance of the machinery and IT investments I will analyze in in Section 3.2.3.

In ASM, plants with more than 250 employees are always included, whereas plants with fewer employees are chosen randomly, with a probability positively correlated with their number of employees. This sampling policy creates bias

Table 4
Creditor rights and productivity: Placebo treatment

Dependent variable: *TFP*

	[1]	[2]	[3]
<i>ARL</i>	0.007 (0.013)	0.008 (0.013)	-0.005 (0.014)
Control variables	Yes	Yes	Yes
Plant fixed effects	Y	Y	Yes
Year fixed effects	Yes	Yes	Yes
State of incorporation fixed effects	No	Yes	Yes
State of headquarters fixed effects	No	Yes	Yes
State of location $\times$ Industry $\times$ Year fixed effects	No	No	Yes
Rounded <i>N</i>	220,000	220,000	220,000
<i>R</i> ²	.74	.74	.77

This table presents estimates of the plant-level effect of placebo antirecharacterization laws on total factor productivity (TFP). In this robustness exercise, I falsely assume that states bordering Texas, Louisiana, or Alabama are treated. The unit of observation in each regression is a plant-year pair. The main independent variable is *ARL*, which is an indicator variable that equals one for firms headquartered or incorporated in falsely treated states between 1997 and 2003. Control variables include the size and age of the plant and the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. * $p < .1$; ** $p < .05$; *** $p < .01$.

toward larger plants. To address this, I reestimate the baseline regression by weighting observations by their respective ASM sample weight. Results in Appendix IA.IX are very similar to those in Table 2.

Finally, I reestimate the baseline regression in Table 2 by allowing time shocks to affect plants of different characteristics differentially. To do this, I interact year dummies with plant size, plant age, and the number of plants belonging to the firm. Estimated coefficients in Appendix IA.X are very similar to those in Table 2, which shows that baseline results are robust to the inclusion of these fixed effects.

3.2 The channel behind creditor rights and increasing productivity

3.2.1 Receivables. From this point on, I investigate the channel through which stronger creditor rights translate into increasing productivity. Antirecharacterization laws apply to all sales of receivables. Therefore, effects of these laws should be more pronounced for firms whose receivables decrease.

I test this hypothesis in Table 5. I classify each firm to be undergoing a change in receivables depending on its lagged change in receivables (*recch*) scaled by lagged assets (*at*). Columns 1 and 2 classify firms below the median of the change in receivables distribution as undergoing a decrease in receivables, and firms above the median as undergoing an increase in receivables.

Estimated coefficients show that effect of these laws is concentrated among plants belonging to firms whose receivables undergo a decrease. This result is in line with the mechanism facilitated by antirecharacterization laws.

3.2.2 Debt. Next, I analyze whether stronger creditor rights as a result of antirecharacterization laws lead firms to increase their borrowing. On the

Table 5
Creditor rights and productivity: Decrease in receivables

Dependent variable: *TFP*

	Decrease in receivables	
	Yes [1]	No [2]
<i>ARL</i>	0.034*** (0.012)	−0.004 (0.016)
Control variables	Yes	Yes
Plant fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
State of incorporation fixed effects	Yes	Yes
State of headquarters fixed effects	Yes	Yes
State of location × Industry × Year fixed effects	Yes	Yes
Rounded <i>N</i>	112,000	83,500
<i>R</i> ²	.84	.84

This table presents estimates of the plant-level effect of antirecharacterization laws on total factor productivity (TFP) across firms undergoing a relative decrease in accounts receivable or not. The unit of observation in each regression is a plant-year pair. The main independent variable is *ARL*, which is an indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2003, or Alabama between 2001 and 2003. Each firm is classified to be undergoing a decrease (increase) in receivables depending on its lagged change in receivables (*recch*) scaled by lagged assets (*at*). Columns 1 and 2 classify firms below the median of the change in receivables distribution as undergoing a decrease in receivables and firms above the median as undergoing an increase in receivables. Plant controls include age and total value of shipments. Control variables include the size and age of the plant and the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. **p* < .1; ***p* < .05; ****p* < .01.

theoretical front, the effect of stronger creditor rights on the borrowing behavior of firms is not obvious. On the one hand, relaxation of collateral constraints would induce firms to borrow more, as predicted in Hart and Moore (1994). On the other hand, Vig (2013) argues that increasing access to collateral by creditors might lead firms to decrease their use of secured debt. Further, leverage also depends on whether or not the originator firm consolidates the SPV onto its balance sheet. In case of consolidation, the balance sheet include both the underlying asset and the borrowing by the SPV. In case of unconsolidation, the originator firm removes the underlying asset from its balance sheet, and does not include the borrowing by the SPV as additional debt in its balance sheet.²⁴

Table 6 shows that treated firms increase their borrowing; this result is consistent with that of Hart and Moore (1994) and Li et al. (2016). In Columns 1 and 2, long-term debt scaled by total assets increases significantly. The estimated coefficient of interest is statistically significant at the 1% confidence level. In Columns 3 and 4, *leverage*, defined as the sum of long-term and short-term debt divided by total assets, increases significantly. In terms of economic magnitude, the estimated coefficient corresponds to approximately 5% increase

²⁴ Consolidation rules regarding SPVs have become stricter over time. In 1990, the Financial Accounting Standards Board (FASB) issued EITF 90-15, which states that firms could avoid SPV consolidation if outside investors maintained equity of more than 3% in a SPV. In 2000, the FASB made reporting gains or losses of any off-balance sheet mandatory. Finally, in 2003, the FASB raised the 3% equity threshold to 10%, making it more difficult to keep SPVs off-balance sheet. See Lemmon et al. (2014) for more details.

Table 6
Creditor rights and productivity: Debt

Dependent variable	Long-term debt		Leverage		Total debt	
	[1]	[2]	[3]	[4]	[5]	[6]
ARL	0.019*** (0.006)	0.018*** (0.006)	0.017*** (0.006)	0.013** (0.006)	0.136** (0.058)	0.137** (0.007)
Sales		0.024*** (0.002)		0.024*** (0.002)		0.024*** (0.002)
Profitability		−0.127*** (0.017)		−0.127*** (0.017)		−0.127*** (0.017)
Tobin's q		−0.009*** (0.001)		−0.009*** (0.001)		−0.009*** (0.001)
Tangibility		0.106*** (0.014)		0.106*** (0.014)		0.106*** (0.014)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rounded N	30,000	30,000	30,000	30,000	30,000	30,000
R ²	.71	.68	.71	.68	.71	.68

This table presents estimates of the firm-level effect of antirecharacterization laws on long-term debt, leverage, and total debt. I define the *Long-term debt*, *Leverage*, and *Total debt* as long-term debt scaled by total assets, sum of long-term and short-term debt scaled by total assets, and the natural logarithm of sum of long-term and short-term debt, respectively. The main independent variable is *ARL*, which is an indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2003, or Alabama between 2001 and 2003. The unit of observation in each regression is a firm-year pair. Firm controls include the natural logarithm of sales, profitability, Tobin's q, and tangibility. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. * $p < .1$; ** $p < .05$; *** $p < .01$.

relative to the sample mean of 0.267. Finally, in Columns 5 and 6, I define the dependent variable as the natural logarithm of the sum of long-term and short-term debt to ensure that the results in Columns 1 to 4 are not driven by the asset values in the denominator. Estimated coefficients are in line with Columns 1 to 4. These results confirm that antirecharacterization laws relax the borrowing constraints of treated firms by allowing creditors more access to collateral.

3.2.3 Composition of investment. To shed further light on the underlying mechanism, in this section, I establish the link between increased borrowing and increased productivity by looking at the composition of plant-level investment and employment.

An important body of literature argues that relaxation of collateral-based lending constraints will result in higher investment by firms.²⁵ However, not all investment is the same in terms of productivity consequences. Firms might be investing in pet projects, or invest in land or buildings that may have little effect on the efficiency of the production process.²⁶ Therefore, the composition of investment plays a critical role for productivity.

²⁵ Chaney et al. (forthcoming), Gan (2007), and Peek and Rosengren (2000) are prominent examples of this literature.

²⁶ Krishnan et al. (2014) argue that firms can easily waste increased access to financing by taking on unproductive or less productive pet projects.

The main reason for relatively limited research on the composition of investment or the composition of a firm's workforce is data limitations. For investment, aggregate items including capital investment expenditures on plant, property, and equipment reported by standard corporate data sources cannot inform us about the vintage of the productive assets in a manufacturing plant. As to employment, difficulty of finding reliable firm- or plant-level data constitutes an additional obstacle against any attempt to study the composition of a firm's workforce.

I focus on two aspects of composition of investment relevant for productivity. The first aspect is vintage of capital. Eisfeldt and Rampini (2007) show that investing in used capital rather than new capital is very common among credit-constrained firms. If new capital is more productive because of technological progress, then firms can become productive by investing in new capital rather than used capital. The second aspect is IT. Research confirms the relationship between IT, computers, and output.²⁷ Two recent studies in the finance literature examine the effect of IT-related investments. First, Brav et al. (2015) show that IT-related investments by hedge funds contribute to the productivity of target firms. Second, Ashwini and Tambe (2016) argue that employees in target firms benefit from IT-related private equity investments by acquiring new skills. They show that workers treated with a private equity investment earn higher wages on average. Therefore, IT-related investment can be a channel through which stronger creditor rights translate into increasing productive efficiency.

First, in panel A of Table 7, I analyze the effect of antirecharacterization laws on the level of investment. Columns 1 and 2 show that *Total investment*, defined as total capital expenditures scaled by lagged capital stock, significantly increases at treated plants. In terms of economic magnitudes, the estimate corresponds to a 7% increase relative to the sample mean, which is significant.²⁸ In Columns 3 to 6, a breakdown of total investment indicates that both machinery and computer investments experience a similar significant increase.^{29,30}

²⁷ By using survey data from 584 establishments and 21 industries, Kelley (1994) shows that computer-controlled machinery is key to efficiency in manufacturing processes. Greenman and Mairesse (1996) examine the French manufacturing and service industries and argue that use of computers influences productivity positively. Furthermore, Black and Lynch (2001), using a nationally representative sample of businesses, show that plant productivity is positively correlated with greater computer usage by nonmanagerial employees. Finally, using plant-level data, Bartel et al. (2007) document that new IT investments make all stages of the production process more efficient.

²⁸ In Appendix IA.XI, I reestimate the baseline model with *TFP* as the dependent variable using the subsample of observations for which machinery investment is not missing. The results are both statistically and economically significant.

²⁹ In Appendix IA.XII, I use the natural logarithm of one plus machinery and computer investments as dependent variables. The results remain qualitatively similar.

³⁰ I use the *capital expenditures for computers* variable from the CMF/ASM databases to proxy for computer and IT-related investments.

Table 7
Creditor rights and productivity: Level and composition of investment

A. Level of investment						
Dependent variable	Total investment		Machinery investment		Computer investment	
	[1]	[2]	[3]	[4]	[5]	[6]
ARL	0.007** (0.003)	0.007* (0.004)	0.013*** (0.004)	0.012** (0.005)	0.0009*** (0.0003)	0.0011*** (0.0004)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of incorporation fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of headquarters fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of location × Industry × Year fixed effects	No	Yes	No	Yes	No	Yes
Rounded N	120,000	120,000	120,000	120,000	90,000	90,000
R ²	.38	.43	.35	.40	.45	.51
B. Composition of investment						
Dependent variable	Fraction new capital		Fraction new machinery-1		Fraction new machinery-2	
	[1]	[2]	[3]	[4]	[5]	[6]
ARL	0.028*** (0.009)	0.024*** (0.007)	0.024*** (0.008)	0.019*** (0.006)	0.030*** (0.007)	0.026*** (0.006)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of incorporation fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of headquarters fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of location × Industry × Year fixed effects	No	Yes	No	Yes	No	Yes
Rounded N	120,000	120,000	120,000	120,000	120,000	90,000
R ²	.37	.42	.36	.42	.34	.47

This table presents estimates of the plant-level effect of antirecharacterization laws on the level (panel A) and the composition of investment (panel B). In panel A, the dependent variables are *Total investment*, *Machinery investment*, and *Computer investment*. *Total investment* is defined as total capital expenditures scaled by lagged capital stock. *Machinery investment* and *Computer investment* are defined as machinery expenditures and computer expenditures scaled by lagged machinery and equipment stock, respectively. In panel B, the dependent variables are *Fraction new capital*, *Fraction new machinery-1*, *Fraction new machinery-2*, and *Fraction computer*. *Fraction new capital* is defined as new capital expenditures over total capital expenditures. *Fraction new machinery-1* is defined as new machinery expenditures over total machinery expenditures. *Fraction new machinery-2* is defined as new machinery expenditures over total capital expenditures. *Fraction computer* is defined as computer expenditures over total capital expenditures. The main independent variable is ARL, which is an indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2001, or Alabama in 2001. The unit of observation in each regression is a plant-year pair. Control variables include size and age of the plant as well as the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. *, $p < .1$; **, $p < .05$; ***, $p < .01$.

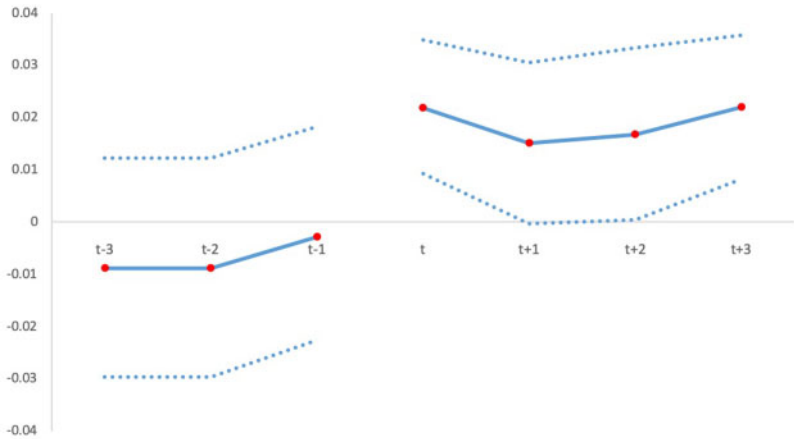


Figure 1
The effect of the adoption of antirecharacterization laws on the “fraction” of new machinery investments
This figure shows the effect of the adoption of antirecharacterization laws on the “fraction” of new machinery investments, defined as new machinery expenditures scaled by total capital expenditures. I estimate Equation (2), except that I replace *ARL* with dummy variables indicating the year relative to the adoption of antirecharacterization laws, where year *t* is the year antirecharacterization law is adopted in a given state. The solid line represents the point estimates associated with each of these dummy variables. The dashed lines represent the 95% confidence interval at which robust standard errors are clustered at the state-of-location level.

Next, I examine the composition of investment in panel B. The disaggregated nature of the Census data allows us to see capital expenditures in new and used goods, separately. In Columns 1 and 2, the dependent variable is *Fraction New Capital*, defined as new capital expenditures over total capital expenditures. I find that the fraction of investment composing new capital significantly increases at treated plants. Estimated coefficients of interest in Columns 1 and 2 are statistically significant at the 1% confidence level, and they indicate an increase of 2.8 and 2.4 percentage points, respectively. This constitutes a significant increase, given that the pretreatment fraction of investment in new capital is 94%.

In the remaining columns of panel B, I analyze how the fraction invested in new machinery and computers is affected. The dependent variable is defined as new machinery expenditures over total machinery expenditures in Columns 3 and 4, and as new machinery expenditures over total capital expenditures in Columns 5 and 6. Estimated coefficients indicate a significant increase in the fraction invested in new machinery. Further, Figure 1 plots the evolution of new machinery expenditures over total capital expenditures over time. The figure is consistent with the “parallel trends” assumption. Finally, Columns 7 and 8 indicate that IT-related expenditures over total capital expenditures significantly increase at treated plants.

In summary, I find strong evidence that not only the level of investment but also the composition of investment change at treated plants. The fraction invested in new capital rather than used capital as well as the fraction invested

in IT increase significantly following the enactment of antirecharacterization laws.

3.2.4 Composition of investment and financial constraints. Next, I analyze how financial constraints affect the composition of investment and productivity. Eisfeldt and Rampini (2007) argue that financial constraints constitute the main reason firms invest in used capital rather than new capital. To reiterate, if new capital is more productive because of technological progress, then firms can become more productive by investing in new capital rather than used capital. This reasoning is in line with Midrigan and Xu (2014), who argue that distortion of technology adoption decisions is an important channel through which financial frictions decrease individual firms' productivity. According to this argument, following the relaxation in borrowing constraints, firms can become more productive by increasing their fraction of investment in new capital.

I use two different measures of financial constraints. I take the median value 1 year before the treatment across all companies to classify firms as financially constrained or financially unconstrained. The first measure of financing constraints is the size-age (SA) index first used in Hadlock and Pierce (2010). I classify firms as financially constrained (unconstrained) if their SA value is above (below) the median. The second measure is the Whited-Wu index first used in Whited and Wu (2006). I classify firms as financially constrained (unconstrained) if their Whited-Wu value is above (below) the median.

Table 8 presents the results. Columns 1 to 3 of panels A and B show that the adoption of antirecharacterization laws has little to no effect on the investment composition of relatively unconstrained firms. Plants belonging to relatively constrained firms, however, experience a significant increase in the fraction invested in new capital and new machinery. Further, changing the composition of investment is associated with higher productivity. Column 4 in both panels shows that plants belonging to relatively constrained firms experience a significant increase in productivity, in line with the predictions of Midrigan and Xu (2014).³¹

3.2.5 Composition of investment and asset redeployability. Next, I analyze how asset redeployability affects the relationship between creditor rights and composition of investment. Eisfeldt and Rampini (2009) argue that because U.S. bankruptcy laws make the repossession of assets easier for lessors than

³¹ Optimality implies that increased access to financing as a result of the relaxation of borrowing constraints will be spread across other units to equate the marginal revenue across units. In Appendix IA.XIII, I test this implication by looking at the productivity of plants in different locations with respect to the state of headquarters and state of incorporation. I classify each plant as *near* or *far* depending on its state of location. I classify a plant as *near* if it is located at either the state of headquarters or the state of incorporation of the firm it belongs to, as *far* otherwise. I see that plants both in and out of these treated states experience an increase in productivity, which is in line with the implication that financially constrained firms spread financing shocks to all of their units.

Table 8
Creditor rights and productivity: Financial constraints

A. Hadlock-Pierce measure

Dependent variable	<i>Fraction new capital</i> [1]	<i>Fraction new machinery-1</i> [2]	<i>Fraction new machinery-2</i> [3]	<i>TFP</i> [4]
<i>ARL</i>	0.006 (0.004)	0.005 (0.004)	0.013 (0.011)	0.002 (0.009)
<i>ARL × Constrained</i>	0.018** (0.008)	0.018** (0.008)	0.051*** (0.018)	0.055*** (0.020)
Control variables	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
State of incorporation fixed effects	Yes	Yes	Yes	Yes
State of headquarters fixed effects	Yes	Yes	Yes	Yes
State of location × Industry × Year fixed effects	Yes	Yes	Yes	Yes
Rounded <i>N</i>	110,000	110,000	110,000	110,000
<i>R</i> ²	.43	.42	.40	.85

B. Whited-Wu measure

Dependent variable	<i>Fraction New capital</i> [1]	<i>Fraction New machinery-1</i> [2]	<i>Fraction New machinery-2</i> [3]	<i>TFP</i> [4]
<i>ARL</i>	0.006 (0.004)	0.005 (0.004)	0.016 (0.011)	0.003 (0.009)
<i>ARL × Constrained</i>	0.020*** (0.008)	0.019** (0.008)	0.030** (0.015)	0.047** (0.019)
Control variables	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
State of incorporation fixed effects	Yes	Yes	Yes	Yes
State of headquarters fixed effects	Yes	Yes	Yes	Yes
State of location × Industry × Year fixed effects	Yes	Yes	Yes	Yes
Rounded <i>N</i>	110,000	110,000	110,000	110,000
<i>R</i> ²	.43	.42	.39	.85

This table presents estimates of the plant-level effect of antirecharacterization laws on composition of investment and total factor productivity (TFP) across ex ante financially constrained and unconstrained firms. *Fraction new capital* is defined as new capital expenditures over total capital expenditures. *Fraction new machinery-1* is defined as new machinery expenditures over total machinery expenditures. *Fraction new machinery-2* is defined as new machinery expenditures over total capital expenditures. The unit of observation in each regression is a plant-year pair. The main independent variable is *ARL*, which is an indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2001, or Alabama in 2001. Panels A and B use the median of Hadlock and Pierce (2010), and Whited and Wu (2006) measures, respectively, to classify firms as constrained or unconstrained. Plant controls include age and total value of shipments. Control variables include the size and age of the plant and the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. * $p < .1$; ** $p < .05$; *** $p < .01$.

for secured lenders, leasing allows for increased debt capacity. This higher debt capacity is valuable, especially for financially constrained borrowers, despite the agency costs associated with leasing. Consistent with this, I show in Section 3.2.4 that after the enactment of antirecharacterization laws allowing secured creditors to seize their assets more quickly during bankruptcy, financially constrained firms switch to new assets from used or leased assets. This changing composition should be more visible for more redeployable assets with an

active secondary-asset markets making the relationship between primary and secondary markets more substitutable (Gavazza 2010; Chu forthcoming).

I use two industry-level redeployability measures, kindly provided by Kim and Kung (2009). To build these industry-level measures, first, a redeployability score for each asset is constructed. For both measures, this score represents the proportion of firms using the asset with the only difference that the second measure also incorporates correlations of outputs among industry peers in the construction of asset redeployability score. The industry-level redeployability measures are then constructed by value-weighting the asset redeployability score of each asset in the industry. I define an industry *redeployable* if its corresponding redeployability measure ranks above the median in a given year.

Table 9 presents the results. First, estimated coefficients on *ARL* are statistically insignificant, which shows that there is no significant change in industries with assets that are less redeployable. Second, plants operating in industries characterized by assets that are more redeployable experience a significant increase in the fraction invested in new capital and new machinery. Estimated coefficients on $ARL \times Redeployable$ are statistically significant at the 1% confidence level. This shows that asset redeployability, which makes the relationship between primary and secondary markets more substitutable, allows firms to switch to new assets following a strengthening of creditor rights in bankruptcy.

3.2.6 Composition of workforce. Next, I analyze the effect of stronger creditor rights on the composition of workforce. According to the capital-skill complementarity hypothesis, equipment capital is more complementary to skilled labor than to unskilled labor (Griliches 1969; Krusell et al. 2000). Therefore, newer capital investment should increase skilled labor's relative productivity, and thereby, its relative demand, which constitutes the hypothesis I test in this section.

I use two new variables from the Census data: *production workers* and *nonproduction personnel*. *Production workers* represent employees (up through the working-foreman level) engaged in production operations at the plant. *Nonproduction personnel* include supervisors (above the foreman level) and white-collar office employees in sales and marketing, financing, purchasing, and professional and technical roles

Panel A of Table 10 shows the growth rate of these different types of labor. Columns 1 and 2 indicate that total number of employees significantly increases at treated plants. This is in line with previous studies arguing that firms hire more after a relaxation in borrowing constraints (Campello and Larrain 2016; Benmelech et al. 2015; Ersahin and Irani forthcoming). A breakdown of aggregate workforce in Columns 3 to 6 shows that growth in nonproduction workers is the driver of growth in the workforce: estimated coefficients of interest for nonproduction personnel are positive and statistically significant at

Table 9
Creditor rights and productivity: Redeployability

Redeployability measure:

Dependent variable	Measure 1			Measure 2		
	Fraction new capital [1]	Fraction new machinery-1 [2]	Fraction new machinery-2 [3]	Fraction new capital [4]	Fraction new machinery-1 [5]	Fraction new machinery-2 [6]
ARL	0.004 (0.005)	0.002 (0.006)	0.009 (0.012)	0.007 (0.006)	0.004 (0.007)	0.013 (0.011)
ARL × Redeployable	0.038*** (0.005)	0.033*** (0.007)	0.032*** (0.011)	0.035*** (0.006)	0.029*** (0.007)	0.025*** (0.009)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of incorporation fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of headquarters fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of location × Industry × Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rounded N	120,000	120,000	120,000	120,000	120,000	120,000
R ²	.42	.42	.39	.42	.42	.39

This table presents estimates of the plant-level effect of antirecharacterization laws on composition of investment across industries varying in redeployability of assets. *Fraction new capital* is defined as new capital expenditures over total capital expenditures. *Fraction new machinery-1* is defined as new machinery expenditures over total machinery expenditures. *Fraction new machinery-2* is defined as new machinery expenditures over total capital expenditures. The unit of observation in each regression is a plant-year pair. The main independent variable is ARL, which is an indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2001 or Alabama in 2001. Industry-level redeployability measures are taken from Kim and Kung (2017). Redeployability measure 1 uses a weighting scheme based on market capitalization of public firms in a given industry-year, whereas Redeployability measure 2 further incorporates correlation of outputs across firms. An industry is considered redeployable if its corresponding redeployability measure ranks above the median in a given year. Control variables include the size and age of the plant and the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 10
Creditor rights and productivity: Workforce

A. Symmetric growth rate

Dependent variable	Total emp.		Production emp.		Nonproduction emp.	
	[1]	[2]	[3]	[4]	[5]	[6]
<i>ARL</i>	0.009*** (0.003)	0.009** (0.004)	0.006 (0.007)	0.002 (0.010)	0.038*** (0.010)	0.041*** (0.012)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of incorporation fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of headquarters fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State of location × Industry × Year fixed effects	No	Yes	No	Yes	No	Yes
Rounded <i>N</i>	108,000	108,000	108,000	108,000	108,000	108,000
<i>R</i> ²	.24	.31	.17	.24	.24	.30

B. Composition of workforce

Dependent variable	Fraction nonproduction emp.	
	[1]	[2]
<i>ARL</i>	0.025* (0.014)	0.027** (0.012)
Control variables	Y	Y
Plant fixed effects	Y	Y
Year fixed effects	Yes	Yes
State of incorporation fixed effects	Yes	Yes
State of headquarters fixed effects	Yes	Yes
State of location × Industry × Year fixed effects	No	Yes
Rounded <i>N</i>	108,000	108,000
<i>R</i> ²	.87	.88

This table presents estimates of the plant-level effect of antirecharacterization laws on growth rate (panel A) and composition of workforce (panel B). In panel A, the dependent variables are symmetric growth rate of *Total employment*, *Production employment*, and *Nonproduction employment*. *Production employment* consist of employees (up through the working foreman level) engaged in production operations at the plant. *Nonproduction employment* consist of supervisors (above the working foreman level) and office employees in sales and marketing, financing, purchasing, and professional and technical roles. In panel B, the dependent variable is *Fraction nonproduction employment*, defined as natural logarithm of number of nonproduction employees over total number of employees. The main independent variable is *ARL*, which is an indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2001 or Alabama in 2001. The unit of observation in each regression is a plant-year pair. Control variables include the size and age of the plant and the number of plants owned by the firm. Table A1 in the appendix defines all variables. Standard errors (in parentheses) are clustered at the state of location level. **p* < .1; ***p* < .05; ****p* < .01.

the 1% confidence level, whereas estimated coefficients for production workers are statistically indistinguishable from zero.

In panel B, I analyze whether this differing growth rate for production and nonproduction workers leads to a change in the composition of workforce. The dependent variable *Fraction nonproduction employment* is defined as the natural logarithm of the number of nonproduction employees over the total number of employees. Column 2 indicates that the share of nonproduction personnel increases by 2.7%. Figure 2 plots the evolution of share of nonproduction employment over time, showing that the increase in the share

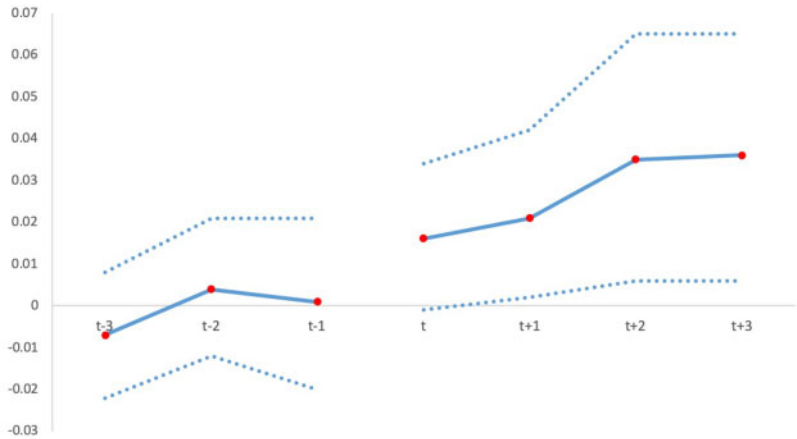


Figure 2
The effect of the adoption of antirecharacterization laws on the “fraction” of nonproduction employment
This figure shows the effect of the adoption of antirecharacterization laws on the “fraction” of nonproduction employment, defined as the natural logarithm of number of nonproduction employees over total number of employees. I estimate Equation (2), except that I replace *ARL* with dummy variables indicating the year relative to the adoption of antirecharacterization laws, where year *t* is the year antirecharacterization law is adopted in a given state. The solid line represents the point estimates associated with each of these dummy variables. The dashed lines represent the 95% confidence interval at which robust standard errors are clustered at the state-of-location level.

of nonproduction personnel is line with the adoption of antirecharacterization laws.

In summary, results are consistent with the capital-skill complementarity hypothesis: treated firms that change their composition of investment also change the composition of their workforce by increasing hiring of skilled labor.

3.3 Comparison to estimates in the literature

My findings are qualitatively similar to previous studies on the effect of collateral laws, despite differences in the magnitudes. Informational asymmetries coupled with the depth of financial markets make collateral laws a greater variation for emerging and transitional economies than for the United States (Haselmann et al. 2010). Using a legal reform in Romania allowing firms to use movable assets as collateral, Campello and Larrain (2016) report dramatic increases in leverage and investment, amounting to 35% and 60% of the average sample leverage and investment ratios, respectively. Further, using cross-country micro-level loan data, Calomiris et al. (2017) show that the loan-to-values of loans collateralized with movable assets are 60.8% in countries with stronger collateral laws. My estimates for leverage and investment nearing 5% and 7%, respectively, are much more moderate when compared with estimates from emerging and transitional economies.

Further, using a legal reform in Sweden that reduced collateral values, Cerqueiro et al. (forthcoming) report similar estimates: a 9% reduction in

total debt and an 8% decrease in investment. Finally, Catherine et al. (2018) use a structural model of firm dynamics with collateral constraints and find that collateral constraints lead to a 2.7% TFP loss in the United States. My reduced-form TFP estimate of 2.6% is strikingly similar to their model-based estimates.

4. Conclusion

Using plant-level data from the Census, I first show that the TFP of plants belonging to treated firms increases significantly, by 2.6%, following the adoption of antirecharacterization laws in Texas, Louisiana, and Alabama. The granularity of the data helps me compare two plants in the same state, industry, and year. The baseline results survive various robustness checks, including a dynamic analysis of the effects of laws and a placebo test where neighboring states are falsely assumed to be treated states.

In the second part of the empirical analysis, I focus on the mechanism, and analyze the composition of investments and workforce. First, I show that treated plants change the composition of their investments by increasing the fraction of capital expenditures in new capital and IT. Next, I show that this effect is seen only among plants belonging to financially constrained firms. This finding is in line with previous research arguing that a relaxation of financial constraints induces firms to increase their productivity by adopting productivity-enhancing projects (Midrigan and Xu 2014; Krishnan et al. 2014). Finally, I provide evidence that treated plants changing the composition of their investments also change the composition of their workforce by hiring more skilled employees. This finding is consistent with theories of capital-skill complementarity (Griliches 1969; Krusell et al. 2000).

In sum, my evidence highlights that creditor rights affect not only the level or quantity but also the composition or quality of a firm's investments and workforce. The relationship between finance and compositional aspects of investments and workforce is a promising avenue for future research.

Appendix A. State Laws

Texas and Louisiana

The section below is from Texas and Louisiana Uniform Commercial Code (UCC).

Section 9-109. Scope

- (e) The application of this chapter to the sale of accounts, chattel paper, payment intangibles, or promissory notes is not to recharacterize that sale as a transaction to secure indebtedness but to protect purchasers of those assets by providing a notice filing system. For all purposes, in the absence of fraud or intentional misrepresentation, the parties' characterization of a transaction as a sale of such assets shall be conclusive that the transaction is a sale and is not a secured transaction and that title, legal and equitable, has passed to the party characterized as the purchaser of those assets regardless of whether

Table A1
Variable definitions

Variable	Definition	Source
<i>A. Plant-level variables</i>		
<i>ARL</i>	Indicator variable that equals one for firms headquartered or incorporated in Texas or Louisiana between 1997 and 2003 or Alabama between 2001 and 2003	
<i>Total factor productivity</i>	Establishment-level log total factor productivity computed following Foster et al. (2016)	CMF/ASM
<i>Labor productivity</i>	Sales minus material and energy costs divided by plant-level total hours	CMF/ASM
<i>Return on capital</i>	Sales minus material and energy costs and payroll divided by plant-level capital stock	CMF/ASM
<i>Capital stock</i>	Sum of structures and equipment calculated using the perpetual inventory method	CMF/ASM
<i>Size</i>	Natural logarithm of the plant's value of shipments	CMF/ASM
<i>Age</i>	Number of years since the first year the plant first appears in the Longitudinal Business Database	LBD
<i>Number of plants</i>	The total number of plants of the parent firm	LBD
<i>Total investment</i>	Total capital expenditures divided by lagged plant-level capital stock	CMF/ASM
<i>Machinery investment</i>	Machinery expenditures divided by lagged machinery and equipment stock	CMF/ASM
<i>Computer investment</i>	Capital expenditures for computers divided by lagged machinery and equipment stock	CMF/ASM
<i>Fraction new capital</i>	New capital expenditures divided by total capital expenditures	CMF/ASM
<i>Fraction new machinery-1</i>	New machinery expenditures divided by total machinery expenditures	CMF/ASM
<i>Fraction new machinery-2</i>	New machinery expenditures divided by total capital expenditures	CMF/ASM
<i>Fraction computer</i>	Capital expenditures for computers divided by total capital expenditures	CMF/ASM
<i>Total employment</i>	Total number of employees working in the plant	CMF/ASM
<i>Production employment</i>	Number of employees (up through the working-foreman level) engaged in production operations at the plant	CMF/ASM
<i>Nonproduction employment</i>	Number of supervisors above the working foreman level and office employees	CMF/ASM
<i>Fraction nonproduction employment</i>	Natural logarithm of nonproduction employees over total number of employees	CMF/ASM
<i>B. Firm-level variables</i>		
<i>Sales</i>	Natural logarithm of sales (<i>sale</i>)	Compustat
<i>Profitability</i>	Operating income before depreciation (<i>oibdp</i>) minus depreciation and amortization (<i>dp</i>) divided by total assets (<i>at</i>)	Compustat
<i>Tobin's q</i>	Number of common stocks (<i>csho</i>) times end of year closing price (<i>prcc</i>) plus total assets (<i>at</i>) minus common equity (<i>ceq</i>) minus deferred taxes (<i>txdb</i>) divided by total assets (<i>at</i>)	Compustat
<i>Tangibility</i>	Property, plant, and equipment (<i>ppent</i>) divided by total assets (<i>at</i>)	Compustat

This appendix presents the definitions for the variables used throughout the paper.

the secured party has any recourse against the debtor, whether the debtor is entitled to any surplus, or any other term of the parties' agreement.

Alabama

The section below are from the 2013 Code of Alabama.

Section 35-10A-1

This chapter may be referred to as the “Asset-Backed Securities Facilitation Act.” It is intended by the Legislature that the term “securitization transaction” be construed broadly.

Section 35-10A-2

- (a) Notwithstanding any other provision of law including, but not limited to, Section 7-9-506 and Section 7-9A-623, to the extent set forth in the transaction documents relating to a securitization transaction:
 - (1) Any property, assets, or rights purported to be transferred, in whole or in part, in the securitization transaction shall be deemed to no longer be the property, assets, or rights of the transferor;
 - (2) A transferor in the securitization transaction, its creditors or, in any insolvency proceeding with respect to the transferor or the transferor’s property, a bankruptcy trustee, receiver, debtor, debtor in possession, or similar person, to the extent the issue is governed by Alabama law, shall have no rights, legal or equitable, whatsoever to reacquire, reclaim, recover, repudiate, disaffirm, redeem, or recharacterize as property of the transferor any property, assets, or rights purported to be transferred, in whole or in part, by the transferor; and
 - (3) In the event of a bankruptcy, receivership, or other insolvency proceeding with respect to the transferor or the transferor’s property, to the extent the issue is governed by Alabama law, such property, assets, and rights shall not be deemed to be part of the transferor’s property, assets, rights, or estate.
- (b) Nothing contained in this chapter shall be deemed to require any securitization transaction to be treated as a sale for federal or state tax purposes or to preclude the treatment of any securitization transaction as debt for federal or state tax purposes or to change any applicable laws relating to the perfection and priority of security or ownership interests of persons other than the transferor, hypothetical lien creditor or, in the event of a bankruptcy, receivership, or other insolvency proceeding with respect to the transferor or its property, a bankruptcy trustee, receiver, debtor, debtor in possession, or similar person

Section 35-10A-3

Any act which becomes effective after September 12, 2001, shall not be construed to amend or repeal any provision of this chapter unless the subsequent act specifically references this chapter and states that this chapter is repealed or states the manner in which this chapter is to be amended. Without limiting the foregoing, Act 2001-481, 2001 Regular Session, does not amend or repeal any provision of this chapter.

Appendix B. Choice of Law Rules of Article 9 Uniform Commercial Code (UCC)

The 1972 and 2002 versions of UCCs are the relevant codes for treatment states. Below is Chapter 3 of Section 9-103 of 1972 Official Text and Comments of Article 9 Secured Transactions.

Section 9-103. Perfection of Security Interests in Multiple State Transactions

- (3) Accounts, general intangibles and mobile goods.
 - (a) This subsection applies to accounts (other than an account described in subsection (5) on minerals) and general intangibles and to goods which are mobile and which are of a type normally used in more than one jurisdiction, such as motor

vehicles, trailers, rolling stock, airplanes, shipping containers, road building and construction machinery and commercial harvesting machinery and the like, if the goods are equipment or are inventory leased or held for lease by the debtor to others, and are not covered by a certificate of title described in subsection (2).

- (b) **The law (including the conflict of laws rules) of the jurisdiction in which the debtor is located governs the perfection and the effect of perfection or nonperfection of the security interest.**
- (c) If, however, the debtor is located in a jurisdiction which is not a part of the United States, and which does not provide for perfection of the security interest by filing or recording in that jurisdiction, the law of the jurisdiction in the United States in which the debtor has its major executive office in the United States governs the perfection and the effect of perfection or nonperfection of the security interest through filing. In the alternative, if the debtor is located in a jurisdiction which is not a part of the United States or Canada and the collateral is accounts or general intangibles for money due or to become due, the security interest may be perfected by notification to the account debtor. As used in the paragraph, "United States" includes its territories and possessions and the Commonwealth of Puerto Rico.
- (d) **A debtor shall be deemed located at this place of business if he has one, at his chief executive office if he has more than ones place of business, otherwise at this residence.** If, however, the debtor is a foreign air carrier under the Federal Aviation Act of 1958, as amended, it shall be deemed located at the designated office of the agent upon whom service of process may be made on behalf of the foreign air carrier.

Below is the official comment (e) for the section above.

- (e) **"Chief executive office" does not mean the place of incorporation; it means the place from which in fact the debtor manages the main part of his business operations.** This is the place where persons dealing with the debtor would normally look for credit information, and is the appropriate place for filing.

As seen above, the 1972 version of UCC defines the *debtor location* to be the location of the Chief Executive Office, which makes the the treatment state to be the state where the headquarters is located. The official comment explicitly states that the Chief Executive Office does not mean the place of incorporation. The 2002 version of UCC changes the location of debtor to be the state of incorporation for registered organizations, as stated below.

Section 9-307. Location of Debtor

- (a) "Place of business."

In this section, "place of business" means a place where a debtor conducts its affairs.

- (b) Debtor's location: general rules.

Except as otherwise provided in this section, the following rules determine a debtor's location:

- (1) A debtor who is an individual is located at the individual's principal residence.
- (2) A debtor that is an organization and has only one place of business is located at its place of business.
- (3) A debtor that is an organization and has more than one place of business is located at its chief executive office.

- (c) Limitation of applicability of subsection (b).

Subsection (b) applies only if a debtor's residence, place of business, or chief executive office, as applicable, is located in a jurisdiction whose law generally requires information concerning the existence of a nonpossessory security interest to be made generally available in a filing, recording, or registration system as a condition or result of the security interest's obtaining priority over the rights of a lien creditor with respect to the collateral. If subsection (b) does not apply, the debtor is located in the District of Columbia.

- (d) Continuation of location: cessation of existence, etc.

A person that ceases to exist, have a residence, or have a place of business continues to be located in the jurisdiction specified by subsections (b) and (c).

- (e) Location of registered organization organized under State law.

A registered organization that is organized under the law of a State is located in that State.

Section 9-102. Definitions and Index of Definitions

- (70) “**Registered organization**” means an organization organized solely under the law of a single State or the United States and as to which the State or the United States must maintain a public record showing the organization to have been organized.

Appendix C. Financial Constraints

I use two measures of financial constraints:

1. **Whited-Wu index:** I follow Whited and Wu (2006) in the construction of the Whited-Wu index. It is constructed as $-0.091 \times ((ib+dp)/at) - 0.062 \times \mathbb{1}(dvc+dvp > 0) + 0.021 \times (dltr/at) - 0.044 \times \log(at) + 0.102 \times (\text{average industry sales growth}) - 0.035 \times \text{sales growth}$, where variables in italic represent the Compustat codes for the variables. Firms above the median are coded as *constrained* and those below the median are coded as *unconstrained*.
2. **Hadlock-Pierce index:** I follow Hadlock and Pierce (2010) in the construction of Hadlock-Pierce index. It is constructed as $-0.737 \times Size + 0.043 \times Size^2 - 0.040 \times Age$, where *Size* is the natural logarithm of total assets (*at*) defined in 2004 dollars and capped at natural logarithm of \$4.5 billion. *Age* is defined to be the number of years the firm is first listed with a nonmissing stock price on Compustat and capped at 37.

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